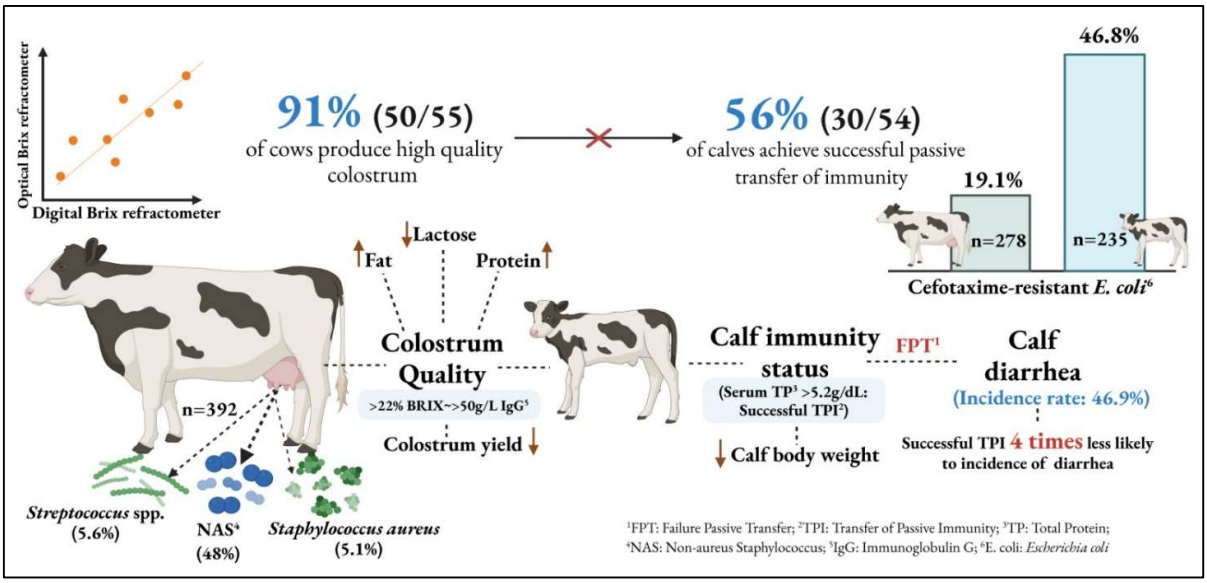
Graphical Abstract



Key highlights of the findings

- **Colostrum quality**: 91% of cows produced good-quality colostrum, but only 56% of calves achieved successful transfer of passive immunity (TPI).
- **Milk composition**: Colostrum quality represented by the optical Brix refractometer was reliable for on-farm colostrum evaluation. Colostrum quality was also positively associated with protein, solid-not fat (SNF), and lactose, supporting their role as complementary indicators of colostrum quality.
- **Risk factors**: Higher colostrum yield reduced quality, and greater calf birth weight reduced success of TPI.
- **Intramammary infection causing pathogens**: Non-aureus staphylococci were the predominant intramammary pathogens (48%).
- **Calf health**: High incidence of calf diarrhea (46.9/100 calves in 15 days) and risk of diarrhoea increased with a lower TPI.
- **Antimicrobial resistance**: Cefotaxime-resistant *Escherichia coli* was more prevalent in calves (46%) than in cows (19%).

Abstract

Newborn calves are born agammaglobulinemic lacking circulating antibodies and rely entirely on colostrum to obtain early immune protection. Therefore, supplementing good quality colostrum is crucial for the successful transfer of passive immunity (TPI) in newborn calves. However, there is limited evidence from Bangladesh evaluating colostrum quality and its impact on TPI in the first month of life, particularly under intensive dairy farming conditions. Therefore, this study aimed to estimate the a) colostrum quality, b) association of colostrum quality with milk composition and udder health parameters, c) underlying cow- and calf-level risk factors impacting colostrum quality and calf health, d) prevalence of intramammary infection (IMI) causing pathogens in composite milk, and e) prevalence of *Escherichia coli* and cefotaxime-resistant *E. coli* between cows and their neonatal calves over time. A five-month cohort study was conducted on an organized commercial dairy farm with fifty-seven cows followed from calving to 30 days postpartum. Composite milk samples (n=392) were collected from each cow on days 1–5, 15, and 30 (day 1 considered colostrum). The composite milk samples were tested using digital and optical Brix refractometers, analyzed for milk composition, somatic cell count, and examined for bacterial identification. Brix values were measured using digital (DR) (MISCO DD-3®, USA) and optical (OR) (ATAGO®, Japan) refractometers. Milk composition (fat, protein, lactose, and solids-not-fat [SNF]) and somatic cell count (SCC) were determined using Ekomilk Horizon Unlimited® and DeLaval Somatic Cell Counter®. Rectal swabs were collected from the cows (n=278) and their calves (n=235) on days 1, 3, 5, 15, and 30 for bacterial culture. Additionally, 2 mL blood samples (n=114) were collected from cows and their newborn calves within 12–24 hours postpartum to assess TPI using calf serum total protein. As data were not normally distributed (Shapiro–Wilk test), pairwise correlations were evaluated using Spearman's rank correlation ($P \leq 0.05$). A strong positive correlation was observed between the digital and optical refractometers across all sampling days ($r = 0.73–0.96$; $P < 0.001$), indicating that either method can be reliably used under field conditions realtime assessment on-farm. On day 1, the 25th percentile of colostrum quality was 12.6% (DR) and 12.0% (OR), while the 75th percentile was 29.9% (DR) and 29.0% (OR). For serum total protein (TP) on day 1, the 25th percentile was 6.6 g/dL in cows and 5.7 g/dL in calves, whereas the 75th percentile was 7.4 g/dL in cows and 6.5 g/dL in calves. Interestingly, fifty cows (50 out of 55;

91%) produced good-quality colostrum (Brix ≥ 22% equivalent to 50g/L IgG); however, only 30 calves (30 out of 54; 56%) achieved successful TPI (calf serum protein > 5.2 g/dL), indicating that underlying factors beyond colostrum quality likely influenced immunoglobulin absorption in the newborn calves. Importantly, the volume of colostrum yield in the 1st milking had a negative impact on colostrum quality (β= -0.2; P < 0.001). In contrast, calf birth weight had a negative impact on success of TPI (β= -0.5; P= 0.004) indicating larger calves might need a higher volume of good quality colostrum than were supplemented. Additionally, a high incidence of calf diarrhea was observed in the selected calves (46.9 cases per 100 calves during the first 15 days counted for a total 608 calf-days at risk). It was found that success of TPI was negatively impacted by the incidence of calf diarrhoea during the follow-up period in the newborn calves (OR= 0.2; P= 0.04). Brix values (DR) were consistently positively related with milk protein (r = 0.39–0.82; P < 0.01) and solids-not-fat (SNF) (r = 0.32–0.85; P ≤ 0.02) across sampling days, and showed strong correlations with lactose from day-2 onwards (r = 0.60–0.66; P < 0.001), suggesting that these milk components may serve as additional indicators of colostrum quality. Overall, 228 composite milk samples were culture-positive, with non-aureus staphylococci comprising 48% of IMI, followed by streptococcus-like organisms, *Streptococcus* spp., *Staphylococcus aureus*, and non-specific gram-negative bacteria. A high prevalence of cefotaxime-resistant *E. coli* in calves (46%) than cows (19%) was detected, suggesting potential risks for early-life bacterial transmission and antimicrobial resistance concerns. In conclusion, this study findings support that routine on-farm colostrum quality testing, administration of adequate colostrum volume adjusted for calf birth weight and strengthened biosecurity measures may help to improve passive immunity transfer and reduce early-life risk for diarrhea in newborn calves.

Keywords: Brix refractometer, transfer of passive immunity, incidence rate, calf diarrhea, intramammary infection, cefotaxime-resistant *Escherichia coli*.

Chapter-1: Introduction

Bovine colostrum is a specialized mammary secretion produced by cows during the pre-parturient period, with the mammary secretions obtained within the first 24 hours after calving widely recognized and defined as colostrum (Jaster, 2005; Marnila and Korhonen, 2011). Milk produced during the subsequent 24–72 hours postpartum is referred to as transition milk, during which the concentrations of immunoglobulins (Ig) and other bioactive components tends to gradually decrease compared to colostrum (Godden et al., 2008; Fahey et al., 2020). Colostrum is an essential source for developing immune protection and enhancing the growth and performance of the newborn calves (Jaster, 2005) . Unlike normal milk, colostrum contains a concentrated source of nutrients and is rich in immunologically and physiologically active components such as Ig (IgG, IgM, and IgA), and of leucocytes, lactoferrin, lysozyme, and cytokines. Among these, IgG is the most abundant one, making up 85% of the total Ig, and is a key constituent of colostrum, alongside the usual milk components (Butler, 1983; Korhonen et al., 2000)

In bovine, newborn calves are nearly agammaglobulinemic therefore, as they don't have transcellular transport of Ig (transcytosis) across placenta to newborn calves. The early neonatal period is therefore particularly critical, because of their susceptibility to infectious diseases leading to morbidity, and mortality. They depend upon the ingestion of colostrum for the adequate transfer of passive immunity (TPI) through intestinal absorption of immunoglobulin (Chase et al., 2008). Feeding of good quality colostrum during the first hours of life is therefore essential as absorption is the greatest at that time due to "open gut" phenomenon which allows big proteins to pass through intestinal epithelial cells and reach the systemic circulation, aiding the newborn dairy calf's ability to absorb Ig from colostrum to provide the highest level of disease resistance (Baumrucker and Bruckmaier, 2014).

Failure of passive transfer of immunity (FPI) may occur when calf is unable to absorb enough Ig from colostrum. Earlier research findings showed that successful TPI largely depends on mainly four key elements: i) feeding colostrum with optimum level of Ig concentration (50g/L), ii) feeding enough quantity of colostrum, iii) feeding the colostrum immediately after birth, and iv) reducing the microbial contamination of colostrum (Weaver et al., 2000; Godden et al., 2008). An adequate volume of high-

quality colostrum must be absorbed within the first hour of life to ensure successful TPI, which has been associated with an improved growth performance and health status of the newborn calves (Hue et al., 2021). In the long term, FPI affects productivity by reducing the milk production in first and second-lactation and resulting increased culling rate of young cows (Faber et al., 2005). It has been reported that calves with FPI are likely linked with an increased risk of infectious diseases from which neonatal calf diarrhea due to *Escherichia (E.) coli* (2.6–45%), rota and coronavirus (17–79%) and *Cryptosporidium parvum* (27–63%) are common cause of morbidity and mortality of calves (Meganck et al., 2014). Along with that, multidrug-resistant (MDR) *E. coli* were detected in both colostrum fed to calves and in calf fecal samples (He et al., 2021), suggesting that contaminated colostrum may serve as a reservoir and transmission vehicle. Studies reported that MDR *E. coli* were more frequently identified in young calves than in adult cattle, indicating that calves may also act as important reservoirs (Jarrige et al., 2020; Oh et al., 2020).

Thus, it is important to assess colostrum quality before feeding it to calves. Colostrum quality can be assessed by IgG content, according to international recommendations satisfactory concentration of colostrum is 50g/L. Colostrum quality can be assessed using various direct or indirect methods. Among which brix refractometer and colostrometer are the most frequently used indirect methods and the radial immunodiffusion assay (RID) is regarded as the most accurate direct reference method of evaluating colostrum quality. Apart from that, assessment of bacterial contamination is also a good indicator of microbiological quality of colostrum. It has been reported that the pathogenic bacterial contamination of colostrum supplemented to the calves disrupts the systemic absorption of IgG molecules in the small intestine (Morrill et al., 2012; Santos et al., 2017). Microbiologically contaminated colostrum can impair calf growth performance and increase morbidity and mortality in farm (Elizondo-Salazar et al., 2010). Additionally, numerous factors are found to be associated with the production, composition and quality of colostrum, as it varies distinctly among individual cows, production type (dairy or beef), breed, parity, dry period length and nutrition, season of the calving (Dunn et al., 2017).

However, in low and middle income countries (LMIC) like Bangladesh, farmers prioritize rearing cows able to produce large volume of milk as a key indicator for farm profitability, often overlooking the importance of colostrum quality on farm

profitability (Pallab and Ghosh, 2021) which may lead to incidence risk of calf mortality and morbidity due to different infectious diseases like diarrhea (25.2%) followed by parasitic infestation (14.8%), anorexia (12.3%), weak calf syndrome (12.7%) and naval ill (5.7%) etc. (Hasan et al., 2017). Therefore, it can be hypothesized that, an effective biosecurity practices of handling colostrum during harvesting and feeding might be crucial for minimizing the transmission of pathogenic bacteria from dams to neonatal calves, as both colostrum and the utensils used for its collection and feeding may act as source of primary transmission. However, in Bangladesh, research on colostrum quality is limited, with available studies primarily focused on variation in colostrum and milk composition in first few days postcalving (Obaidullah et al., 2008; Akter et al., 2020). There is a clear lack of data describing the time-dependent changes in colostrum quality, and its impact on calf health during the early life of the neonates. To meet these knowledge gaps, this study aimed to estimate the a) colostrum quality, b) association of colostrum quality with milk composition and udder health parameters, c) underlying cow- and calf-level risk factors impacting colostrum quality and calf health, d) prevalence of intramammary infection (IMI) causing pathogens in composite milk, and e) prevalence of *Escherichia coli* and cefotaxime-resistant *E. coli* between cows and their neonatal calves over time.

Chapter-2: Review of Literature

This chapter start with an overview of the dairy industry in Bangladesh; the challenges faced in dairy production highlighting an important area such as colostrum quality in dairy cows and calf mortality. The review then elaborates the biological and management factors affecting calf health, with a particular focus on the role of colostrum quality in neonatal immunity and survival. This includes the physiology of colostrum production, its immunological and nutritional properties, and the principles of effective colostrum management, including feeding timing, quantity, quality, and standard practices. The chapter further discuss the methods for assessing colostrum quality, factors influencing its quality, and the impact of udder health and microbiological contamination on transfer of passive immunity (TPI). The chapter ends by noting important knowledge gaps identified from the existing pertinent scientific evidence and highlighting areas for future research.

2.1 Importance of dairy industries in Bangladesh

Livestock is a cornerstone of Bangladesh's agricultural economy, engaging approximately 80% of the rural population in cattle farming (Ahmed et al., 2023) and directly contributing 1.9% to the national gross domestic product (GDP), which represents 16.5% of the total agricultural GDP (Hossain et al., 2022). According to the report of Bangladesh Dairy Farmers Association, Bangladesh has approximately 1.2 million dairy farms, engaging approximately 9.4 million individuals in direct or indirect employment within the dairy sector (VetEffect, 2023).

Among livestock products, milk ranks second only to meat and eggs as a primary source of animal protein. Despite a recommended daily intake of 250 mL of milk per person, actual per capita availability of milk remains lower (FAO, 2025) , reflecting a persistent gap between supply and demand. Each year, per capita growth rate of milk production is 3.6%, which is still lower than the per capita demand of milk (4%) (Uddin et al., 2011; FAO, 2025) indicating a large gap between requirement and availability of milk. Although both cattle numbers and milk production have increased in recent years, the annual per capita growth rate of milk remains below the per capita demand, underscoring ongoing challenges in meeting national requirements (DLS, 2022). As developing countries now account for 53% of global milk production, a figure projected to rise to 59% over the next decade (OECD/FAO, 2025). Bangladesh's dairy sector is

undergoing rapid transformation, this shift is characterized by a move from smallholders to more intensive, commercial dairy farming with larger, genetically improved herds and advanced management practices (Uddin et al., 2020; VetEffect, 2023). However, productivity gains are constrained by herd health management challenges, importantly those associated with calf mortality. In low- and middle-income countries (LMIC) like Bangladesh, calves are often neglected due to limited immediate economic return, leading to increased risks of morbidity and mortality as they are mainly kept facilitating milk let-down in cows. Nevertheless, the efficiency of dairy farms mainly rely on the survival of calves (Samad, 2001). As the calf mortality poses severe economic and production challenges that significantly affect the sustainability of dairy farming in Bangladesh. Dairy producers can avert economic losses, enhance production efficiency, and contribute to a more sustainable dairy production landscape in Bangladesh by reducing calf mortality.

2.2 Economic impact of calf mortality in dairy production

Calf mortality poses a significant economic challenge for dairy farms, resulting in direct and indirect non-negligible financial losses, reduced genetic advancement, and increased veterinary costs (Abdelhadi et al., 2018). Direct economic losses from calf mortality include substantial replacement costs. In the U.S., annual damages reach approximately $125 million due to stillbirth and calf mortality rates (Raboisson et al., 2013). The monetary loss stems not only from the direct valuation of the calf but also includes potential future milk production losses, which further amplifies the economic detriment experienced by dairy farmers. Indirect losses encompass reduced herd productivity, increased veterinary expenses, and diminished genetic quality of future replacements, often exceeding direct losses (Compton et al., 2017). Although specific figures for indirect costs per herd are variable, they can be estimated to range significantly depending on operational scale and management protocols (Molla et al., 2017). In Bangladesh, calf mortality remains a challenge within the dairy sector. Nationally, the mortality rate is equal to 12%, with notable regional and seasonal variation, highest in Sylhet and Khulna, and peaking during the monsoon (Parvez et al., 2020). Male calves and neonates are especially vulnerable, with most deaths occurring in the first month of life, accounting for 35.2% of total mortality (Islam et al., 2005). Delays or deficiencies in colostrum intake, poor colostrum quality, substandard colostrum and calf management, combined with poor calf-related hygiene practices

such as cleaning routine for feeding equipment, calving pens, calf feeder, can have significant effect on calf health and mortality (Godden et al., 2008; Uetake, 2013; Renaud et al., 2018).

2.3 What explains the dependence of calves on colostrum for early immunity?

In cattle and other ruminants, the placenta is syndesmochorial or epitheliochorial, which creates a complete barrier between maternal and fetal circulatory systems, thereby preventing prenatal transfer of immune components (Blum and Baumrucker, 2008). Consequently, calves are born agammaglobulinemic and depend entirely on colostrum ingestion for immunoglobulin (Ig) acquisition and the establishment of early systemic and mucosal immunity (Calloway et al., 2002). Although neonatal calves have all major immune system components, these are immature and present at significantly lower levels than in adults until 5–8 months of age (Chase et al., 2008). At birth, circulating B-cell numbers are about one-third of adult levels, and key T-cell subsets also do not reach full maturity until several months postnatally, leaving calves immunologically vulnerable (Outteridge, 1985). Neutrophils, while abundant in neonates, exhibit limited phagocytic capacity compared to adults (Menge et al., 1998). This overall underscores the critical importance of TPI via colostrum to provide immediate protection to the neonatal calves.

2.4 Definition of colostrum and transitional milk

The term "colostrum" originates from the Latin "colostrum" or "beestings," referring to the initial mammary secretion produced at calving. The significance of colostrum feeding for calf survival was first recognized in 1937, as colostrum contains high concentrations of immune-active components and macro- and micronutrients vital for the early development in neonatal calves (Hanson, 1961; Weaver et al., 2000; Lopez and Heinrichs, 2022). In dairy cattle, colostrum is defined by its Ig concentration, typically at least 5%, with the highest levels of Ig and lactoferrin present in the first two milkings. Transitional milk, produced after colostrum, retains residual immune-active components and differs nutritionally from mature milk (Weaver et al., 2000). The duration and definition of colostrum vary among studies, with some defining as the first milking in first 24 hours (Lopez-Bondarchuk and Costa, 2026), while some considering 3–5 days of milking postcalving as colostrum (Zarcula et al., 2010; McGrath et al., 2016; Poonia and Shiva, 2022). In this review, colostrum refers to first milking

collected within the first 24 hours postcalving, milk produced during the subsequent 24–72 hours (2 to 3 days of postpartum) is classified as transitional milk, whereas milk produced thereafter is defined as mature milk (Van Soest et al., 2020; Abdelsattar et al., 2022; Lopez-Bondarchuk et al., 2026).

2.5 Applications of bovine colostrum in animal and human health

Bovine colostrum has long been recognized for its antimicrobial, immunomodulatory, and gut-protective properties, with applications extending from traditional medicine to modern sciences. Increasing scientific evidence supports its role in enhancing immune function, improving gut health, and mitigating infectious and inflammatory conditions in both humans and animals, highlighting its potential as a functional food and therapeutic adjunct (Otto et al., 2011; Jones et al., 2019; Lopez et al., 2022).

Key applications of colostrum in animal health

- Essential for transfer of passive immunity (TPI); colostrum replacers are useful when maternal colostrum is unavailable but cannot replace high-quality colostrum (Godden et al., 2008; Lopez et al., 2022).
- Supports gut development, immune function, and disease resistance in calves, piglets, foals, and small ruminants (Fenger et al., 2014; Fenger et al., 2016; Nordi et al., 2016).
- Improves growth performance, feed efficiency, and survival in poultry and pigs (Afzal et al., 2017; Playford and Weiser, 2021).
- Enhances gut morphology, antioxidant activity, and immune responses in fish (Falahatkar et al., 2014; Moretti et al., 2019).
- Improves gut health, immune responses, and vaccine efficacy in companion animals (dogs and cats) (Satyaraj et al., 2013; Gore et al., 2021).

Key applications of colostrum in human health

- Exhibits antibacterial, antiviral, and anti-inflammatory properties (Jones et al., 2019).
- Enhances gut barrier integrity and promotes growth of beneficial microbiota (e.g., *Bifidobacterium* spp., *Lactobacillus* spp.) (Jones et al., 2019; Davison, 2021).
- Reduces severity and duration of viral and bacterial diarrhea, including *E. coli* and rotavirus (Otto et al., 2011).
- Improves immune function and reduces upper respiratory tract infections, particularly in athletes (Jones et al., 2019).

- Attenuates nonsteroidal anti-inflammatory drugs (NSAIDs) and chemotherapy-induced gastrointestinal injury (Pontoppidan et al., 2015; Arslan et al., 2021).
- Shows therapeutic benefits in short bowel syndrome and inflammatory bowel conditions (Filipescu et al., 2018; Spalinger et al., 2019).

Overall, bovine colostrum represents a biologically valuable resource attributing to human and animal health, with growing relevance for functional nutrition, disease prevention, and sustainable livestock and health management systems.

2.6 Colostrogenesis

Colostrogenesis is mainly ante-partum transfer of Ig into mammary secretions, which is district, fast, starts several weeks before calving and appears to be abruptly ceased at calving. The origin and translocation of different classes of Ig into colostrum may follow distinct pathways. Immunoglobulins A and M originate predominantly from local synthesis by plasmacytes of the mammary alveolar cells, after which they are subsequently secreted directly into the developing colostrum (Larson et al., 1980). In this prepartum stage, IgG1 are predominantly and selectively transferred from maternal circulation into mammary secretions within 3–7 weeks prior to calving or during the final weeks of gestation. Although IgG1 and IgG2 appear in similar proportions in the bovine serum, colostrum is heavily enriched in IgG1, approximately 5–10 times higher than IgG2, due to vesicular transport mechanisms of IgG1 within mammary epithelial cells. The small amount of IgG2 found in bovine colostrum is thought to originate from passive movement across the mammary epithelium, entering the secretion through transiently permeable tight junctions in the blood–milk barrier (Samarütel et al., 2016). Endosomal transfer of mammary IgG1 can occur through binding to the receptor complex called neonatal crystallizable fragment receptor (FcRn). After binding to its specific Fc receptor on the basolateral surface of mammary epithelial cells, IgG1 is internalized and transported across the cell via vesicular pathways, ultimately being released into the developing colostrum at the apical cell surface (Liebler-Tenorio et al., 2002). Colostral IgG concentration varies widely among cows, reflecting individual differences in the timing and rate of transfer into the gland, but the physiological factors governing these differences remain only partly understood. Endocrine changes and alterations may regulate the endosomal trafficking within mammary epithelial cells or the expression of FcRn is regulated by hormones such as estrogen, progesterone, and prolactin (Barrington et al., 1997). Colostrogenesis, also involves the buildup of

numerous other colostral constituents that support neonatal growth, immunity, and gastrointestinal development. Primary colostrum components, including Ig, other proteins, fat and lactose, progressively accumulate in the final one to two weeks of gestation and reach maximal concentrations around calving. However, how other bioactive components such as growth factors, cytokines accumulate prepartum are limited, despite the substantial cow-to-cow variability observed in their concentrations (McGrath et al., 2016). A clearer understanding of how these compounds are deposited in the mammary gland before calving is needed to develop strategies aimed at improving colostrum quality for newborn calves.

2.7 Colostrum composition

Colostrum is typically a thick, yellow, slightly acidic secretion (pH ~ 6.4) containing a wide range of biologically active and bacteriostatic components (Table 2.2). It delivers essential nutrients, vitamins, and energy that support neonatal survival during the first hours of life (McGrath et al., 2016) (Table 2.1). The chemical composition of colostrum varies among animal species (Table 2.3) and changes rapidly after parturition, with a progressive decline in both nutritional and biological value over time (Table 2.4). Timely feeding is therefore critical, as optimal absorption occurs when colostrum is ingested immediately after birth to maximize its nutritional and immunological benefits (Figure 2.1).

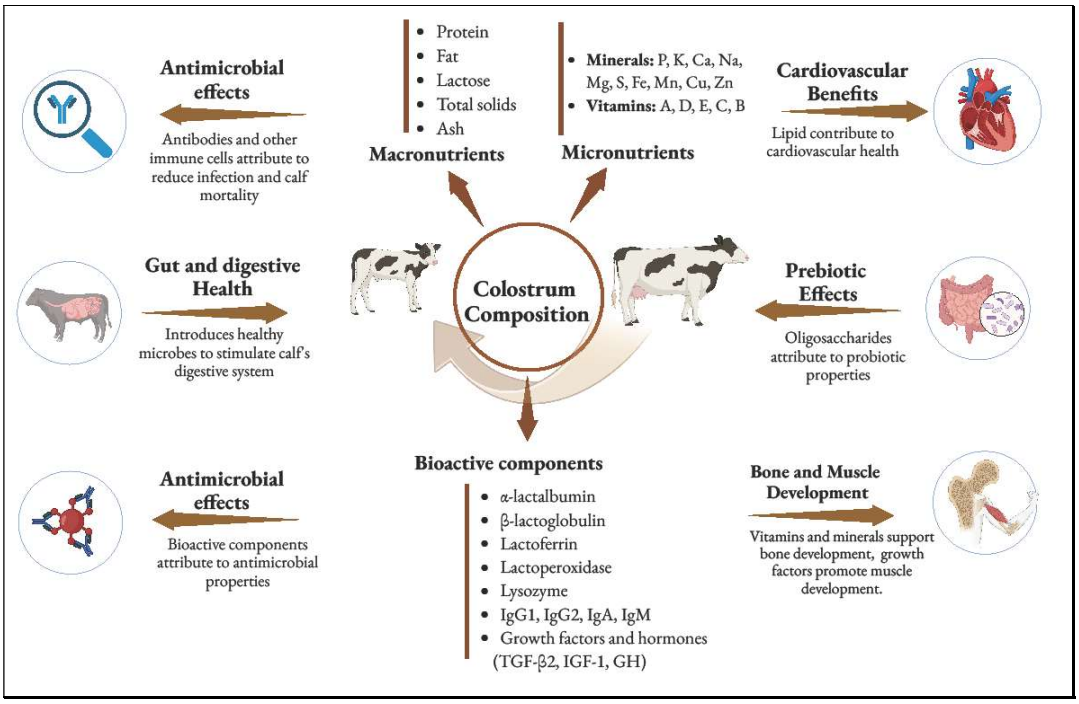


Figure 2.1 An overview of the nutritional and bioactive components of bovine colostrum with their impact in animal and human health (Ig: Immunoglobulin; TGF-β2: Transforming Growth Factor β2; IGF-1: Insulin-like Growth Factor 1; GF: Growth Factor).

Most studies on bovine colostrum composition rely on single-time-point sampling, providing limited insight into its dynamic transition to milk. Few studies in Bangladesh reported that during the first 5 d postpartum, most colostrum constituents declined rapidly toward milk levels, while lactose and pH increased; rear quarters produced colostrum richer in fat and protein than forequarters (Obaidullah et al., 2008; Akter et al., 2020). More comprehensive evaluations are essential to understand the proper transition dynamics from colostrum to milk over time.

Table 2.1 Major bioactive components of bovine colostrum and their biological functions (Miranda et al., 2023).

Bioactive components	Biological functions
Fat	Regulation of body temperature and provision of energy
Lactose	Regulation of body temperature, provision of energy, and stimulation of the nervous system
Protein (Immunoglobulins)	Immune response
Other protein (casein and albumin)	Muscle development and energy
Antimicrobial factors	Immune response

Growth factors	Muscle development and energy
Maternal leucocyte	Immune response
Hormones	Growth performance
Vitamins	Immune response and antioxidant
Nutrients	Muscle development and energy
Cytokines	Immune and anti-inflammatory response
Enzymes (lysozyme, lactoferrin and lactoperoxidase)	Antimicrobial response
Minerals	Growth performance

2.7.1 Proteins

Bovine colostrum is characterized by a higher protein concentration (approximately 15% w/w) compared with milk (about 3%) (Sats et al., 2020). Immunoglobulins constitute approximately 70–80% of total colostral protein, with IgG accounting for 85–90% of total Ig, while IgA and IgM contribute about 5% and 7%, respectively. There are two subtypes of IgG (IgG1 and IgG2), IgG1 predominates, representing approximately 80–90% of total IgG (Larson et al., 1980; Kehoe and Heinrichs, 2007). Immunoglobulin A supports pathogen neutralization by preventing microbial adherence to epithelial surfaces. Immunoglobulin M, the largest isotype, is highly effective at bacterial agglutination, complement activation, and opsonization (Pike, 1967; Tsuruta et al., 2012). The IgG is the primary determinant of TPI in neonatal calves as IgG constitutes the dominant fraction of colostral Ig and is absorbed most efficiently across the intestinal epithelium (Newby et al., 1982). In addition to Ig, colostrum contains proteins such as lactoferrin, transferrin, albumin, and α-lactalbumin, which support nutrient transport, antimicrobial defense, and lactose synthesis (Kehoe et al., 2007).

2.7.2 Carbohydrates

Colostrum has relatively low lactose (2.5%) content than milk (5%) but includes small quantities of other carbohydrates such as glucose, fructose, glucosamine, galactosamine, and various oligosaccharides (Gopal and Gill, 2000; McGrath et al., 2016). Martín et al. (2002) reported that colostrum (0.7–1.2 g/L) contains higher concentrations of oligosaccharides than in milk (0.1 g/L). Colostral oligosaccharides (3-sialyllactose, 6- sialyllactose, 6-sialyl-N-acetyllactosamine) also contribute as an additional layer of protection by preventing pathogens from attaching to intestinal epithelial surfaces (Przybylska et al., 2007). Colostral oligosaccharides exert prebiotic

effects by serving as metabolic substrates for beneficial bacteria such as *Bifidobacterium*, which are key early colonizers of the calf gut (Bondue et al., 2016).

2.7.3 Fat

Fat in colostrum is a critical component of the neonatal calf's diet, providing essential energy for thermoregulation and serving as a precursor for the synthesis of key enzymes and hormones. Fat concentrations in colostrum typically ranging from 4.5% to 7.9% primarily in the form of milk-fat globules (McGee et al., 2006). Compared with milk, colostrum has lower concentrations of trans-fatty acids and short-chain fatty acids (C4–C10) (Contarini et al., 2014). The fatty acid profile of colostrum is composed of about 65–75% saturated fatty acids, 24–28% monounsaturated fatty acids, and 4–5% polyunsaturated fatty acids (Yang et al., 2015; O'Callaghan et al., 2020; Wilms et al., 2022).

2.7.4 Vitamins and minerals

Among vitamins, colostrum constitutes higher concentrations of fat-soluble vitamins (A, D, and E) and carotenoids, as well as water-soluble vitamins including thiamine, riboflavin, B6, B12, folate, and vitamin C. Their concentrations may vary according to the cow body reserves, diet, and season, etc. (Kehoe et al., 2007) . As these vitamins do not efficiently cross the placenta, newborn calves rely almost entirely on colostrum to achieve adequate levels after birth (Quigley and Drewry, 1998). Colostrum is also rich in essential minerals such as calcium, sodium, copper, selenium, iron, zinc, magnesium, manganese, and phosphorus all of which contribute to early metabolic function and tissue development (McGrath et al., 2016). These micronutrients support fat and protein metabolism, provide antioxidant protection, and contribute to proper nervous system function (Buttar et al., 2017).

2.7.5 Cytokines and immune regulators

Cytokines are a diverse group of proteins, peptides, and glycoproteins secreted primarily by T-helper cells and activated macrophages. Although present in colostrum at very low concentrations (10–1,000 pg/mL), they exert anti-inflammatory effects that contribute to immune activation and the recruitment of defenses against pathogenic bacteria, viruses, and fungi (Playford et al., 2021).

MicroRNAs (miRNAs) with immune-regulatory potential are also present in colostrum (Sun et al., 2013). These short, noncoding RNA molecules modulate gene expression at the post-transcriptional level and may serve as a mechanism of postnatal signaling from dam to offspring. Packaged within microvesicles, miRNA remain stable during gastrointestinal transit and can influence lymphoid cell activity in the intestine (Van Hese et al., 2020).

2.7.6 Growth factors and other nutrients

Growth-related bioactive molecules such as transforming growth factor-β2, growth hormone, insulin, and particularly insulin-like growth factor-I (IGF-I and IGF-2) are present markedly in colostrum (Pakkanen and Aalto, 1997). Insulin-like growth factor-I is thought to play an important role in early gastrointestinal development by promoting mucosal growth, enhancing brush-border enzyme activity, stimulating intestinal DNA synthesis, increasing villus height, and improving glucose uptake (Baumrucker et al., 1994; Bird et al., 1996; Bühler et al., 1998). Several colostral enzymes contribute important functions after ingestion, including antioxidant enzymes such as lactoperoxidase, which exhibits potent antimicrobial activity (McGrath et al., 2016).

2.7.7 Maternal leucocytes

Bovine colostrum contains more than one million immunologically active maternal leukocytes per milliliter, comprising macrophages, T and B-lymphocytes, and neutrophils (Larson et al., 1980; Le Jan, 1996). Some colostral leukocytes can be absorbed intact through the neonatal intestinal barrier, entering systemic circulation to enhance T-cell activation and support antibody production. These maternal leukocytes rapidly migrate from the bloodstream into lymphoid and non-lymphoid tissues including mesenteric lymph nodes, Peyer's patches, and the spleen within 24 hours of colostrum intake (Aldridge et al., 1998). Colostrum is heavily enriched with macrophages, which typically account for 50% to 90% all colostral leukocytes (Ellis et al., 1996). A considerable proportion of these maternal macrophages are believed to migrate into the peripheral circulation of newborn calves shortly after colostrum intake.

Table 2.2 An overview of major nutritional, minerals, and immunological components in bovine colostrum and milk (Puppel et al., 2019; Playford et al., 2021; Linehan et al., 2023).

Component	Nutrients	Bovine colostrum	Milk
Macro-nutrients (%)	Total solids	24-28	12.9
	Fat	6-7	3.6-4.0
	Protein	14-16	3.1-3.2
	Casein	4.8	2.5-2.6
	Albumin	6.0	0.4-0.5
	Lactose	2-3	4.7-5.0
Micro-nutrients (g/kg)	Zinc	11.6-38.1	3.0-6.0
	Calcium	2.6-4.7	1.2-1.3
	Phosphorus	4.5	0.9-1.2
	Potassium	1.4-2.8	1.5-1.7
	Sodium	0.7-1.1	0.4
	Magnesium	0.4-0.7	0.1
Immunoglobulins (Ig) (g/L)	Total immunoglobulins	42-90	0.4-0.9
	IgG1	34.0-87.0	0.31-0.40
	IgG2	1.6-6.0	0.03-0.08
	IgA	3.2-6.2	0.04-0.06
	IgM	3.7-6.1	0.03-0.06
Vitamins	Thiamin (B1) (µg/mL)	0.58-0.90	0.4-0.5
	Riboflavin(B2) (µg/mL)	4.55-4.83	1.5-1.7
	Niacin (B3) (µg/mL)	0.34-0.96	0.8-0.9
	Cobalamin(B12) (µg/mL)	0.05-0.60	0.004-0.006
	Vitamin A (µg/mL)	25	34
	Vitamin D (IU/g fat)	0.89-1.81	0.41
	Tocopherol (E) (µg/g)	2.92-5.63	0.06
Proteins	Lactoferrin (g/L)	1.5-5	0.02-0.75
	Lactoperoxidase (mg/L)	11-45	13-30
	Lysozyme (mg/L)	0.14-0.7	0.07-0.6
Amino acids (g/kg)	Aspartic Acid	42.95	28.83
	Threonine	33.26	14.64
	Serine	44.95	19.74
	Glutamic Acid	88.84	91.12
	Proline	25.96	56.98
	Cystine	8.51	2.39
	Glycine	15.65	5.96
	Alanine	15.79	11.42
	Valine	28.33	16.95
	Methionine	9.31	12.00
	Isoleucine	15.1	13.67
	Leucine	47.30	35.94
	Tyrosine	39.56	15.34
	Phenylalanine	25.22	17.16
	Histidine	14.60	12.12

	Lysine	40.90	28.51
	Arginine	14.40	10.22
Growth factors (ng/mL)	IGF-I [1]	100-2,000	5-100
	IGF-2	150-600	5-100
	TGF- b1 [2]	1-50	<5
	TGF- b2	150-1,150	10-70
	EGF[3]	4-325	1-150
Carbohydrates (g/L)	3-sialyllactose	0.42–1.24	0.13
	6-sialyllactose	0.05-0.13	0.04
	6-sialyl-N-acetyllactosamine	0.07-0.42	0.01
Fatty acids (g/100 g)	SFA[4]	66.12-74.05	64.88-64.47
	MUFA[5]	24.54-30.09	31.51-31.79
	PUFA[6]	3.79-3.88	3.61-3.74

[1]IGF: Insulin-like growth factor; [2]TGF: Transforming growth factor; [3]EGF: Epidermal growth factor; [4]SFAs: Saturated fatty acids; [5]MUFAs: Monounsaturated fatty acids; [6]PUFAs: Polyunsaturated fatty acids

Table 2.3 Comparative inter-species variation of macronutrient composition of colostrum (fat, protein, and lactose; g/L) (Puppel et al., 2019).

Species	Fat	Protein	Lactose
Cattle	36	130	31
Goat	90	80	25
Sheep	124	130	31
Horse	7	191	46
Pig	72	180	24

Table 2.4 Temporal changes in the major compositional components of bovine colostrum (%) during the first 120 hours postcalving, representing the rapid transition from colostrum to milk following calving (Puppel et al., 2019).

Time after birth (hours)	Protein	Casein	Albumin	Fat	Lactose
0	16.8	4.1	12.7	6.7	2.9
6	11.7	3.5	8.0	6.1	3.5
12	6.3	3.1	3.2	4.4	3.9
24	5.5	2.9	2.6	4.1	4.1
48	4.8	2.8	2.0	3.9	4.2
120	3.6	2.7	0.9	0.8	4.5

2.8 Principles of colostrum management

Effective colostrum management is commonly summarized by the "three Qs": quickness, quantity, and quality together with cleanliness of colostrum harvesting and feeding (Figure 2.2). It emphasizes the timely provision of adequate volumes of high-quality colostrum to ensure successful TPI and optimal calf health and productivity

(Hunt, 2024). Poor colostrum management has been identified as a major contributor to excessive calf losses in dairy systems (Wells et al., 1996).

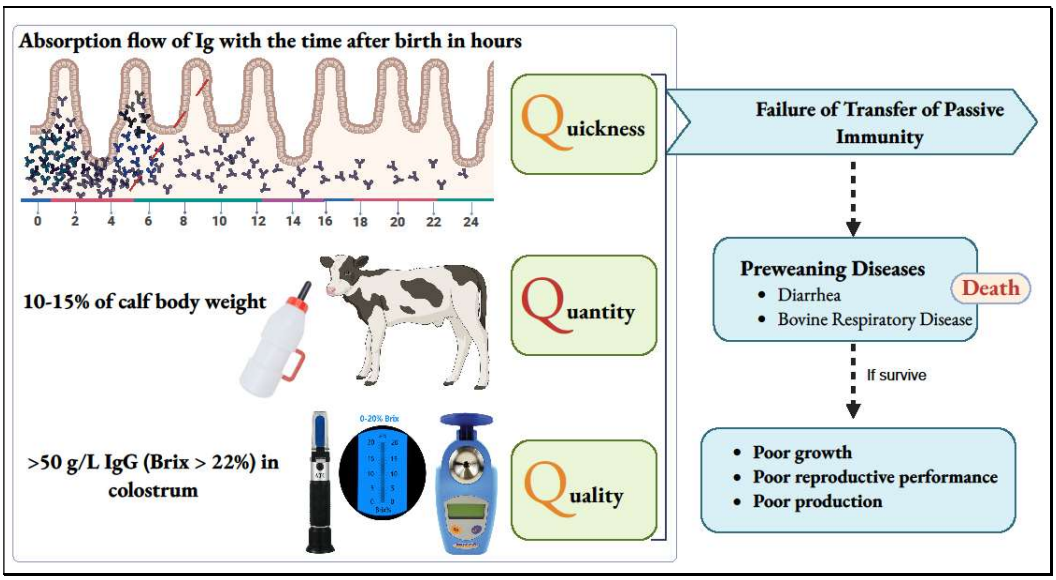


Figure 2.2 The three Qs of colostrum management (Quickness, Quantity, and Quality) and their impact on transfer of passive immunity and calf health outcomes.

2.8.1 Quickness: Timing of feeding

There is a law in Europe under Directive 2008/119/EC on minimum standards of protection of calves where mentioned that "every calf shall receive bovine colostrum as soon as possible after birth and in case first six hours of life" (Río, 2021). Colostral IgG concentration declines by approximately 3.7% per hour after calving, making early first milking the most critical determinant of colostrum quality (Morin et al., 2010). The timing of first colostrum feeding is critical, as neonatal calves rapidly lose the ability to absorb Ig after birth. During the brief "open gut" period, intestinal enterocytes non-selectively absorb intact Ig via pinocytosis, which are subsequently transported into the circulation. Intestinal "closure" occurs progressively, with Ig absorption capacity decreasing by approximately one-third by 6 hours, two-thirds by 12 hours, and becoming negligible by 24 hours postpartum (Table 2.5) ; therefore, colostrum should be fed as soon as possible after birth, ideally within the first 2–4 hours (Zarcula et al., 2010; McGrath et al., 2016).

Table 2.5 Temporal decline in immunoglobulin absorption efficiency and corresponding colostrum feeding recommendations in neonatal claves (Hunt, 2024).

Time after birth (in hours)	Efficiency of absorption (%)	Recommended feeding volume or quantity (% of Body weight)
<1	Peak absorption (90-100%)	10%
1-2	Very high (70-90%)	10%
2-6	High (50-70%)	10%
6-12	Moderate (30-50%)	10%
12-24	Low (10-30%)	10%

2.8.2 Quantity of colostrum fed

The quantity of colostrum fed at the first feeding is equally a key factor. General recommendations suggest providing a volume equivalent to at least 10%–12% of the calf body weight at first feeding (Godden et al., 2008) which means calf should be fed approximately 3 to 4 liters of colostrum at birth, followed by an additional 2 liters within 12 hours after birth. Morin et al. (1997) reported that calves fed 4 liters of colostrum at birth and 2 liters in12 hours interval have higher blood serum IgG levels at 24 hours of life. It is recommended that large-breed calves requiring approximately 4L and smaller-breed calves requiring about 3L of colostrum at first feeding immediately after birth. Calves receiving ≥4 L of high-quality colostrum show reduced morbidity and mortality and improved calf serum IgG (Hunt, 2024).

2.8.3 Quality of colostrum

The concentration of IgG has historically been used to define colostrum quality, with high-quality colostrum containing ≥50 g of IgG/L (Godden et al., 2008). However, colostrum quality is also substantially influenced by several modifiable management factors, including prepartum vaccination strategies, adequate dry period length, and timely collection of colostrum after calving, all of which can be optimized at the farm level to improve TPI in calves.

2.8.4 Cleanliness of colostrum

The fourth golden rule, *cleanliness and storage of colostrum*, is vital for effective management (Table 2.6). The udder should be cleaned before milking, and colostrum promptly transferred to clean containers and rapidly cooled at 1–4 °C for up to 24 hours for short-term storage. Strict hygiene, proper equipment sanitation, and staff training in hygienic handling and temperature management are essential to maintain optimum